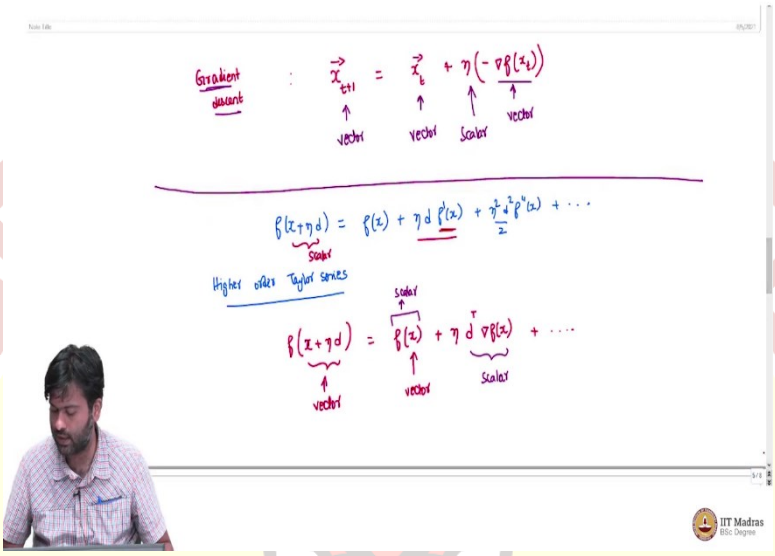


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Taylor Series in Higher Dimensions

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So now, what we are going to do is, we are going to quickly see, just make sure what happens when you look at the Taylor series in the higher dimension. We argued that when you had just 1 dimension, the Taylor series kind of told us why - f dash of x is the thing to do as a choice for the direction. Now, what we are going to see is what is the analog of a Taylor series in the, analogous Taylor series for higher dimensions.

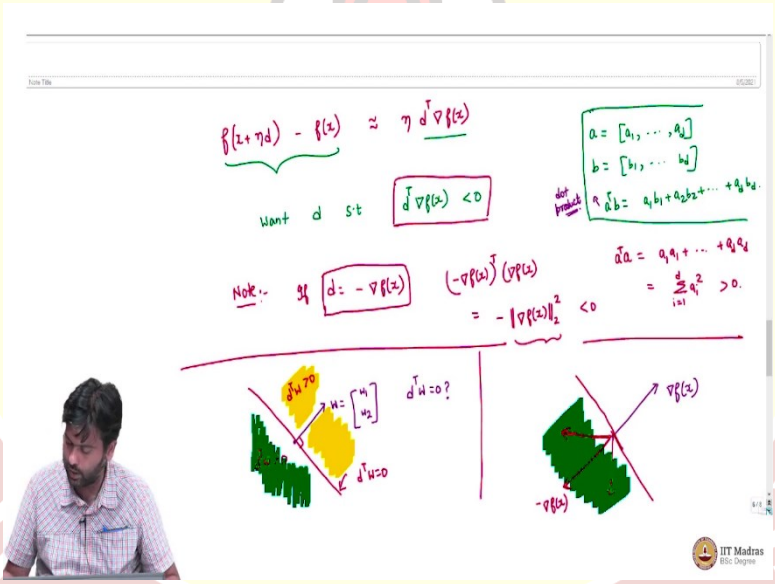
And we will see why that is also a useful thing to look at. So, so what was Taylor series earlier, so it was f of x was simply, well rather you wanted to evaluate f at a different point f of x + some eta d and we said that it is, it can be got by as by evaluating the function at x + eta d f dash of x + eta squared d squared f double dash of x by 2, and so on.

Now, the higher order analog, higher order Taylor series looks like the following. Well, again, you are going to evaluate the function at x + eta d. Now, the x + eta d is actually a vector, here it was a scalar because it was a 1 dimensional function, now it is a higher dimensional function, but nothing much changes. So, this is still going to be f of x. Again, x is a vector. Remember, I mean, just want to make sure the objects are clear though we are using the same notation. +, now this you cannot simply multiply vectors like this.

Now, what happens in the higher order analog is that, well, the eta stays. Now d is a vector f dash the vector, now what we do is we look at d transpose δf of x. Well, we know that the generalization of f of f dash of x is δf of x. Now, D is a vector now, we take the, the directional derivative along d, which is d transpose δf of x. So, this d transpose δf of x is a scalar. Again, f of x is a scalar, x is a vector, you are asking what is the function value at a particular vector, that is a scalar.

Eta is a scalar; d transpose δ f of x is a scalar. And of course, there are these higher order terms. Let us ignore the higher order terms for now. So, this, these there are these other higher order terms. Now, this is the Taylor series expansion for a function for with multiple variables. Now, what, what do we gain by looking at this, well, we gained the following. So now, we will again ask the question, what is a good d to choose? So, the argument is exactly the same. But then let me go through it quickly. So that we understand the important point we will come out of it.

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So, we are again asking, so if f of x + eta d - f of x is this is what we care about. We know that by Taylor series approximation, this is something like eta × d transpose δf of x, of course, I am sure all of you already know what is a transpose b. So, if you have 2 vectors a, which is a 1 till ad and b which is b1 till bd. Now, a transpose b will be something like a 1, b 1 + a 2, b 2 ...+ ad bd, so just to make sure we are on the same page. So, which means that I am multiplying the corresponding component of d with δf of x and then adding them up.

So, now, again, we want to choose a d such that this value is negative, the left hand side is negative. So, want d such that left hand side is negative equivalently we want d such that eta is a positive scalar. We do not care about that; we will just say we want d such that d transpose

δf of x < 0. Now, this would be the corresponding, corresponding equation that we want our direction to satisfy at any given x.

Now, of course, if so now note that if d = - δf of x which is what my gradient descent does, then this becomes - δf of x transpose δf of x, which is what we will call as - norm δf of x squared. Well, the norm, remember the norm of any vector with itself, norm squared is a transpose a, which is a 1 × a 1 + ...+ ad × ad, which is sum over i is 1 to d, ai squared. And this is always a positive quantity, so this is positive, and then I have a negative term here.

So, I know that this is always less than 0. So, if I choose the = - δof x, then I am guaranteed that if I take a small enough step in the direction of t, then my function will decrease. It is the same analogous thing that we already saw. But the interesting thing is that, well, if you look at it, all we are saying is that as long as you have a d, such that d transpose δf of x < 0, you are good. So, of course, if d = - δf of x, the negative of the gradient, then that is a good direction to move, because your function is going to decrease, but that is not the only direction.

So now there can be other ds such that d transpose δf of x < 0, in which if you move, you will still reduce your function. Now, where are these directions, so this is what we are going to ask next. Maybe I will write this here. So where are the ds such that d transpose δ x < 0? Well, the way to think of this is as follows. Let us say I am at some point, let us say we are in 2 dimension.

And there is a vector w that points in this direction. Let us say, some arbitrary W for now. Which is W 1, W 2, think of this as 1 , 1, if you will. Now, I am asking the question, where are the ds such that d transpose W = 0? And we will see why this is useful. So, where are the ds of course, d is a 2 dimensional vector, which are the 2 dimensional vectors which will make 0 when you do a, what is called as a .product, so this is called as a .product. I am sure again, you must have seen this earlier in the course.

So where are the d such that d transpose W = 0, well, these will exactly be those points on the straight line. So, which is, which are in 2 dimension, which are what are called as perpendicular to the W vector. So, any point on this line, you take any point on this line, it satisfies d transpose W = 0. Now, the question then is, where are those guys for whom, for where are those ds for which d transpose W = d transpose W > 0.

Well, this line is d transpose W = to 0. Now, the d is where d transpose W will be greater than 0 will precisely be the guys on this side. So, the 1s shaded in yellow. So, these are d transpose

W > 0. Basically, it will make a positive .product with W. And the guys which for whom will have d transpose W less than 0 are on the other side.

So, this is d transpose W < 0. So, which means given a W it kind of partitions the entire 2 dimensional space into 2 parts, in 1 part, you have d transpose W > 0 and in other part, you have d transpose W < 0, the set of d such that d transpose W < 0 is the green set and the analogous one for the greater than 0 is the yellow set.

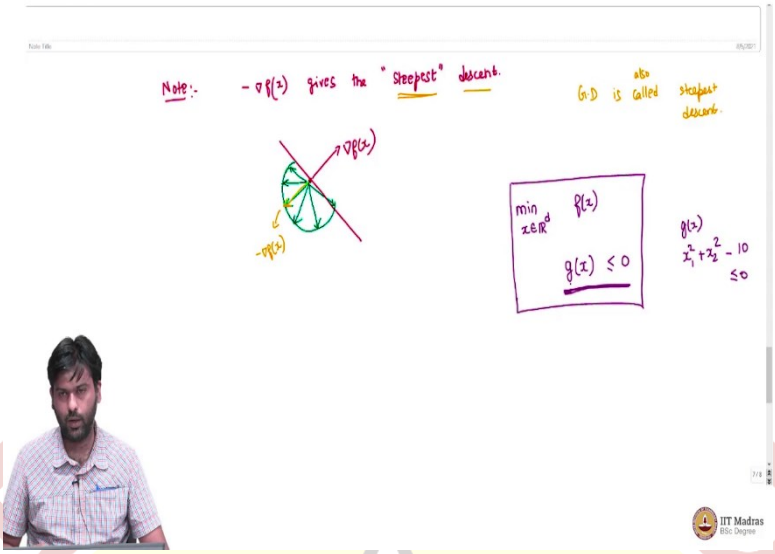
What does this imply for our problem, well, for our problem, we have, we are at some point, let us say x and now, we are looking at well where is the gradient of the function pointing at this at this point, let us say the gradient is pointing in this direction δf of x and we are looking at the gradient and then we want to decide which direction should I move, so that the functions value actually decreases if we take a small step along the direction, that is the question we are asking.

Now, what is this telling us this picture that we just saw tells us that well the direction that you need to move so that you are function value actually decreases is, is on this side, so it is, so you can move anywhere along the green direction, maybe this is a possible direction, maybe this is a possible direction, this is a possible direction there are infinite directions here, if you take a small step in any of these directions, your function value is guaranteed to reduce, for a small enough step size.

Now, the point is the gradient descent algorithm picks a special direction, which is exactly opposite to δf of x. So, which is - δf of x, this is the gradient descent direction, but then that does not mean that is the only direction where you will descend, you can potentially move in a different direction, and also reduce your function, you can potentially descend in any of these directions.

Now, then the question comes, why should I care about the negative gradient direction, I could potential, there may be a different direction, which might be better, so that is the immediate question because now you have infinite directions to move in. And which one should I pick, well, it turns out that and maybe I will write it down different page.

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As a separate note, - δf of x, there is no .here; gives the steepest descent. So, if you are allowed to move only a particular unit length, let us say, so you are at this point, and your gradient is pointing in this direction.

And if you are allowed to move only one unit in whichever direction you want to move, and you have several directions to move, so you can move 1 unit in whichever direction of course, there is no point in moving in any direction, which is, which does not make a negative .product with a gradient, because that is not going to be descent direction, but then, if we, if we are allowed to move in any of these directions, the one that will give you the maximum decrease in the function is can be shown to be the negative gradient direction.

So, this is - δf of x, and this is the direction of steepest descent. So, this gradient descent algorithm is also called some× the steepest descent algorithm, also called the steepest descent. So, the gradient itself is going to point to the direction in which the function is increasing as much as possible, the maximum the direction where the function is increasing the maximum and, and so the negative of the gradient is the direction in which the function is decreasing the maximum.

And so, this function is called this, this algorithm is also some× called the steepest descent algorithm. Well, that does not give us any new insights per se. So, other than saying that, well, we initially started off with this negative gradient. Now, what are we saying, we are saying that, well, that is the best thing to do, in some sense, because you will get them most descent in your function if you move along the negative of the gradient.

So, are there any situations where moving in a different direction other than the negative of the gradient would even make sense? Well, the answer is yes and no. The answer is really not, is no, when you have an unconstrained optimization problem, like the one we are solving now, so you want to just minimize f of x. So that is the problem we have been looking at, at least, so far where you want to, if you have an x, maybe the x is in some d dimension, there is no constraint on x.

And if you want to minimize f of x, then gradient descent is the right thing to do. So, the negative of the gradient is the right direction to move. But if you have constraints, so if you now have, let us say, a constraint, and let us, for simplicity, look at the case where you have only one constraint, and this is an inequality constraint. We will talk about the other cases in a bit. But let us say now we have this problem.

So now, you want to minimize f of x the same function you want to minimize, but then your x cannot be any possible x, it is constrained by the fact that g of x < or equal to 0. Remember, we had this the rope constraint, which was constraining where the cow could move, so it was, in that case, if you remember, g of x was something like x x 1 squared + x 2 squared was - some I think it was 10 < or equal to 0, where 10 was the radius of the rope.

So, this was a constraint. So not all x's are allowed, we are only, we are only allowed to use x from a certain range. In this case, that is indicated by the fact that g of x < or equal to 0. Now, it becomes a very interesting problem, because now, if you move in the negative gradient direction, it could so happen that you will end up in a point for which g of x less than or equal to 0 is no longer satisfied, then, how do we solve this problem?

So, we do want to solve a problem such that g of x < or equal to 0 is satisfied, but then we still want to minimize a f of x. So now, what can we do? So, now, in that particular case, do other directions help us somehow? So, instead of moving in the gradient direction, is there a different direction or what is the deal? So, how can we be sure that we are moving in the right direction or not?

So, now, this is a constrained optimization problem, which is what we initially wanted to solve. Right now, we have solved the unconstrained version of it, the moment you introduce constraints, you already see that there are going to be issues with the gradient descent algorithm. And we will start talking about constraint optimization problems specifically with a single inequality constraint going forward, maybe I will stop here for the moment.